Dashboard Example: Virtual Machine Performance

Cloud Insights

Tony Lavoie
September 22, 2020

This PDF was generated from https://docs.netapp.com/us-en/cloudinsights/task_dashboard_example_VM_performance.html on July 21, 2021. Always check docs.netapp.com for the latest.

Table of Contents

Dashboard Example: Virtual Machine Performance 1

Dashboard Example: Virtual Machine Performance

There are many challenges facing IT operations today. Administrators are being asked to do more with less, and having full visibility into your dynamic data centers is a must. In this example, we will show you how to create a dashboard with widgets that give you operational insights into the virtual machine (VM) performance in your environment. By following this example, and creating widgets to target your own specific needs, you can do things like visualizing backend storage performance compared to frontend virtual machine performance, or viewing VM latency versus I/O demand.

About this task

Here we will create a Virtual Machine Performance dashboard containing the following:

- a table listing VM names and performance data
- a chart comparing VM Latency to Storage Latency
- a chart showing Read, Write and Total IOPS for VMs
- a chart showing Max Throughput for your VMs

This is just a basic example. You can customize your dashboard to highlight and compare any performance data you choose, in order to target for your own operational best practices.

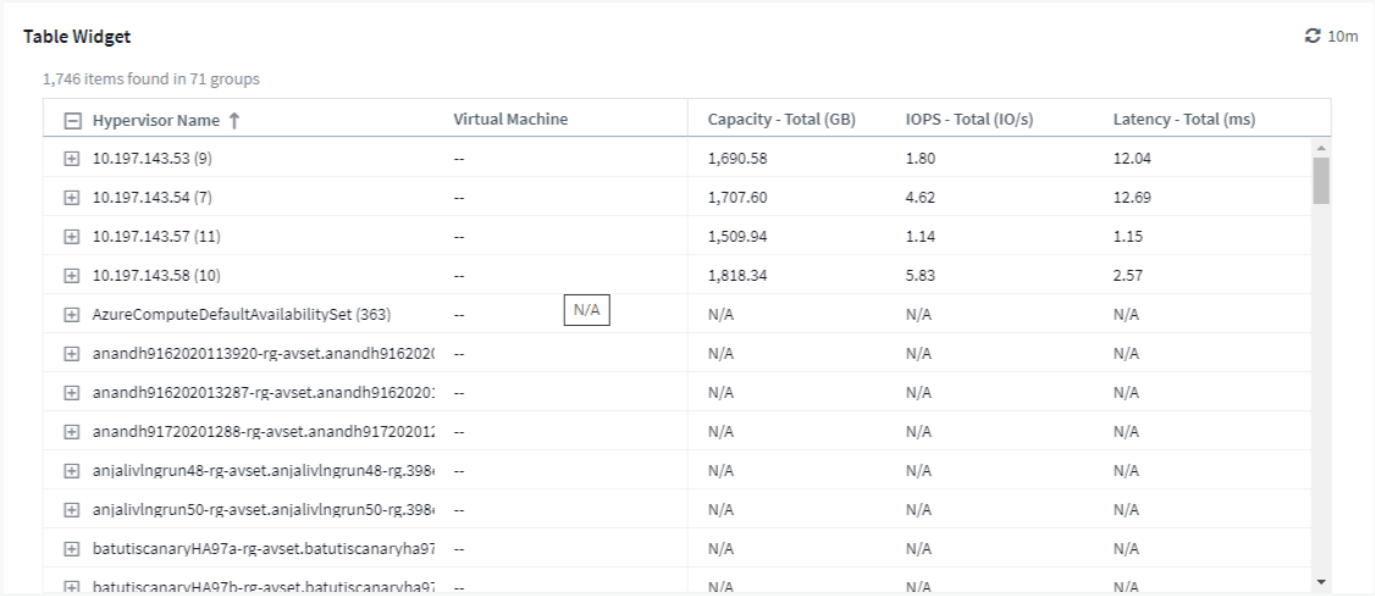
Steps

1. Log in to Insight as a user with administrative permissions.
2. From the **Dashboards** menu, select **[+New dashboard]**.

   The **New dashboard** page opens.

3. At the top of the page, enter a unique name for the dashboard, for example "VM Performance by Application".
4. Click **Save** to save the dashboard with the new name.
5. Let’s start adding our widgets. If necessary, click the **Edit** icon to enable Edit mode.
6. Click the **Add Widget** icon and select **Table** to add a new table widget to the dashboard.

   The Edit Widget dialog opens. The default data displayed is for all storages in your environment.



1. We can customize this widget. In the Name field at the top, delete "Widget 1" and enter "Virtual Machine Performance table".
2. Click the asset type drop-down and change *Storage* to *Virtual Machine*.

   The table data changes to show all virtual machines in your environment.

3. Let's add a few columns to the table. Click the Gear icon on the right and select *Hypervisor name*, *IOPS - Total*, and *Latency - Total*. You can also try typing the name into the search to quickly display the desired field.

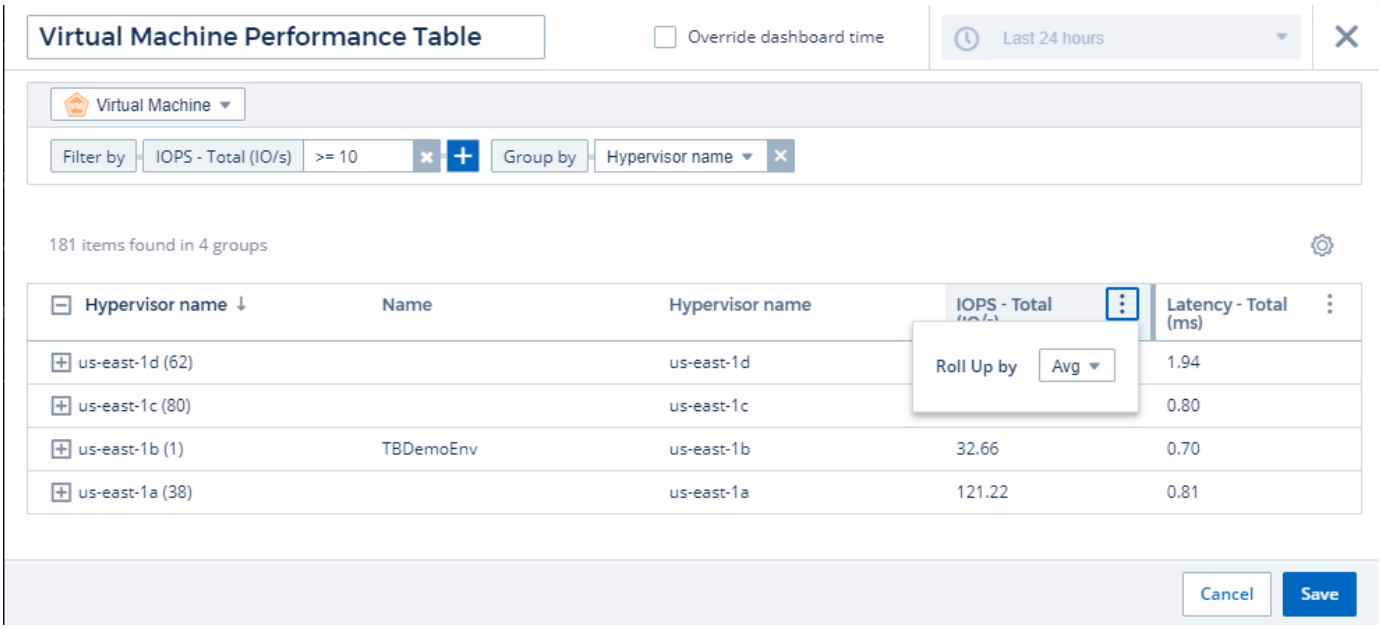
   These columns are now displayed in the table. You can sort the table by any of these columns. Note that the columns are displayed in the order in which they were added to the widget.

4. For this exercise we will exclude VMs that are not actively in use, so let's filter out anything with less than 10 total IOPS. Click the **[+]** button next to **Filter by** and select *IOPS - Total*. Click on **Any** and enter "10" in the **from** field. Leave the **to** field empty. Click outsude the filter field or press Enter to set the filter.

   The table now shows only VMs with 10 or more total IOPS.

5. We can further collapse the table by grouping results. Click the **[+]** button next to **Group by** and select a field to group by, such as *Application* or *Hypervisor name*. Grouping is automatically applied.

   The table rows are now grouped according to your setting. You can expand and collapse the groups as needed. Grouped rows show rolled up data for each of the columns. Some columns allow you to choose the roll up method for that column.



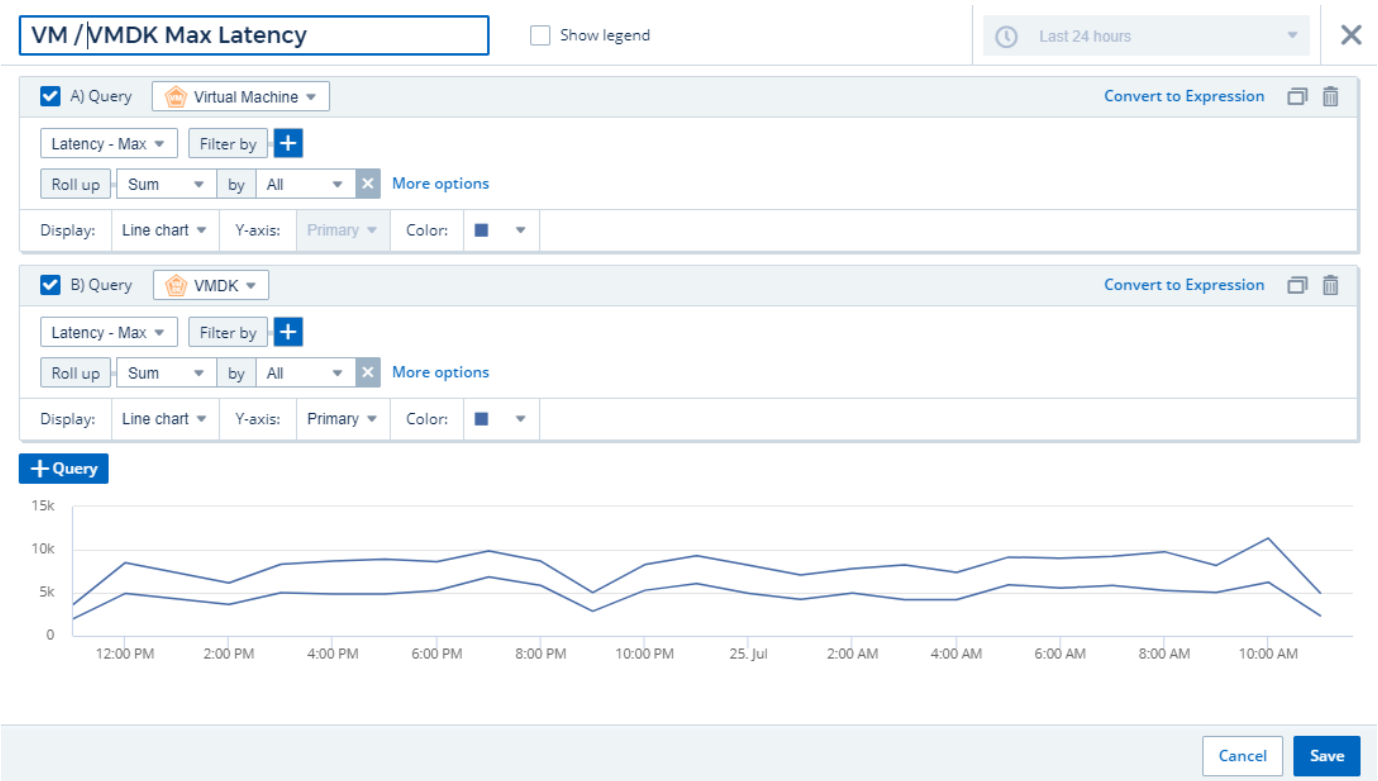
1. When you have customized the table widget to your satisfaction, click the **[Save]** button.

   The table widget is saved to the dashboard.

You can resize the widget on the dashboard by dragging the lower-right corner. Make the widget wider to show all the columns clearly. Click **Save** to save the current dashboard.

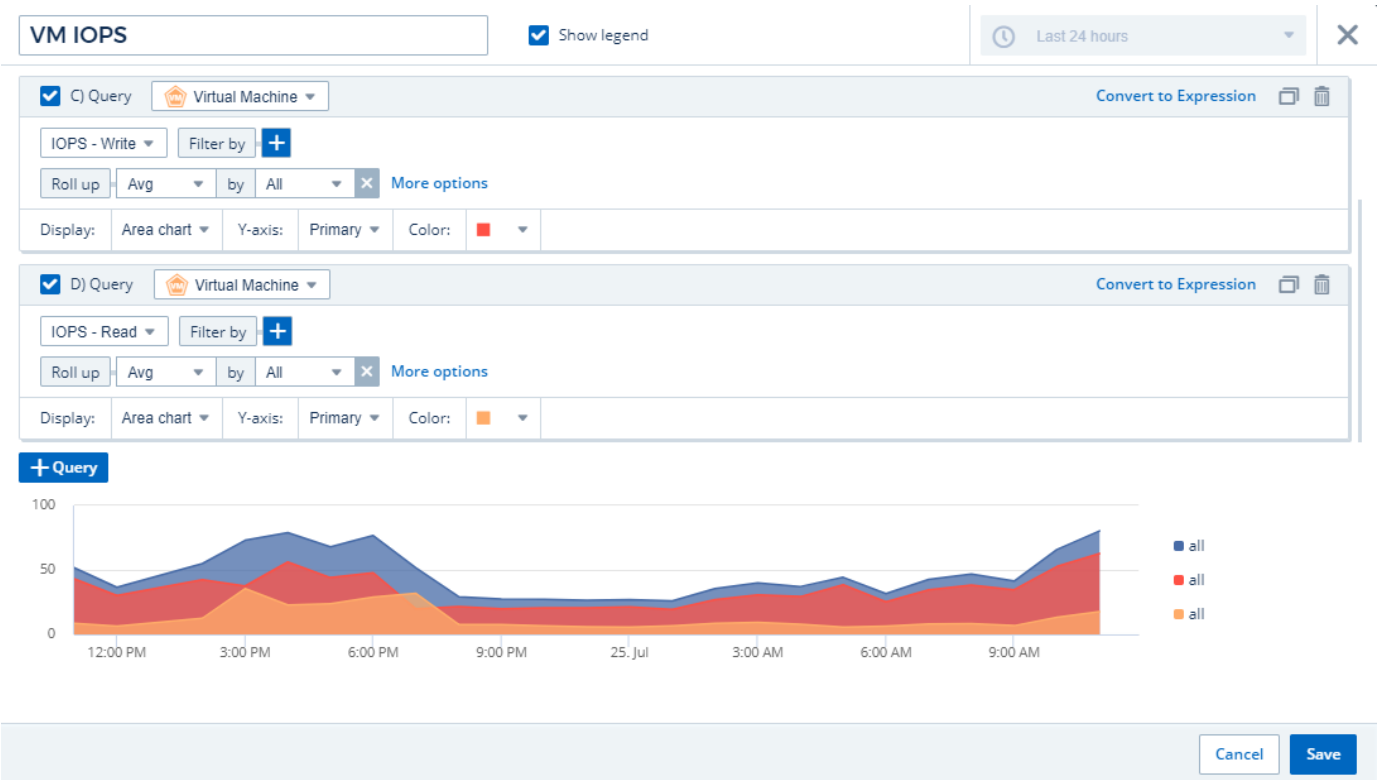
Next we will add some charts to show our VM Performance. Let's create a line chart comparing VM latency with VMDK latency.

1. If necessary, click the **Edit** icon on the dashboard to enable Edit mode.
2. Click the **[Add widget]** icon and select *Line Chart* to add a new line chart widget to the dashboard.
3. The **Edit Widget** dialog opens. Name this widget "VM / VMDK Max Latency"
4. Select **Virtual Machine** and choose *Latency - Max*. Set any filters you wish, or leave **Filter by** empty. For **Roll up**, choose *Sum* by *All*. Display this data as a *Line Chart*, and leave *Y-Axis* as *Primary*.
5. Click the **[+Query]** button to add a second data line. For this line, select *VMDK* and *Latency - Max*. Set any filters you wish, or leave **Filter by** empty. For **Roll up**, choose *Sum* by *All*. Display this data as a *Line Chart*, and leave *Y-Axis* as *Primary*.
6. Click **[Save]** to add this widget to the dashboard.



Next we will add a chart showing VM Read, Write and Total IOPS in a single chart.

1. Click the **[Add widget]** icon and select *Area Chart* to add a new area chart widget to the dashboard.
2. The Edit Widget dialog opens. Name this widget "VM IOPS"
3. Select **Virtual Machine** and choose *IOPS - Total*. Set any filters you wish, or leave **Filter by** empty. for **Roll up**, choose *Sum* by *All*. Display this data as an *Area Chart*, and leave *Y-Axis* as *Primary*.
4. Click the **[+Query]** button to add a second data line. For this line, select **Virtual Machine** and choose *IOPS - Read*.
5. Click the **[+Query]** button to add a third data line. For this line, select **Virtual Machine** and choose *IOPS - Write*.
6. Click **Show legend** to display a legend for this widget on the dashboard.



1. Click **[Save]** to add this widget to the dashboard.

Next we will add a chart showing VM Throughput for each Application associated with the VM. We will use the Roll Up feature for this.

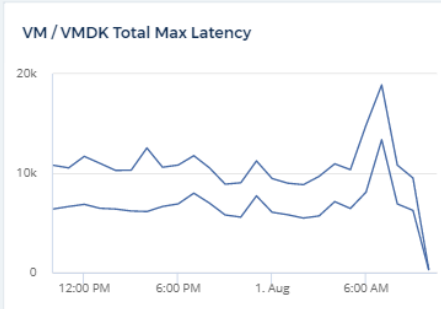
1. Click the **[Add widget]** icon and select *Line Chart* to add a new line chart widget to the dashboard.
2. The Edit Widget dialog opens. Name this widget "VM Throughput by Application"
3. Select Virtual Machine and choose Throughput - Total. Set any filters you wish, or leave Filter by empty. For Roll up, choose "Max" and select by "Application" or "Name". Show the Top 10 applications. Display this data as a Line Chart, and leave Y-Axis as Primary.
4. Click **[Save]** to add this widget to the dashboard.

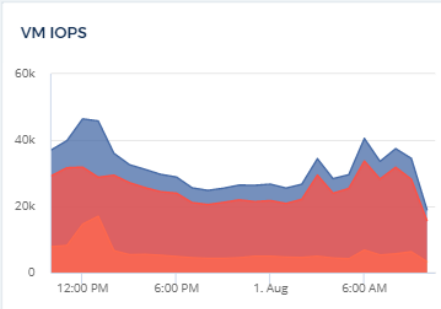
You can move widgets on the dashboard by holding down the mouse button anywhere in the top of the widget and dragging it to a new location.

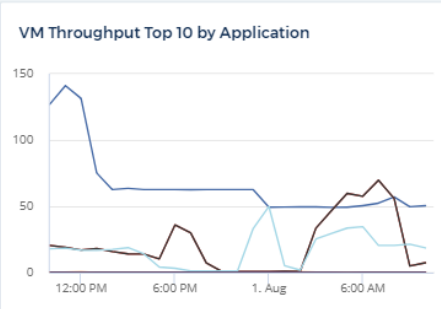
You can resize widgets by dragging the lower-right corner.

Be sure to **[Save]** the dashboard after you make your changes.

Your final VM Performance Dashboard will look something like this:







VM Performance Table

146 items found in 3 groups

Hypervisor name	Name ↑	Hypervisor name	IOPS - Total (IO/s)	Latency - Total (ms)
us-east-1c (65)		us-east-1c	185.84	2.04
us-east-1d (51)		us-east-1d	90.00	1.81
us-east-1a (30)		us-east-1a	86.84	1.00

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.